Lecture 9

Digital Electronics Basics

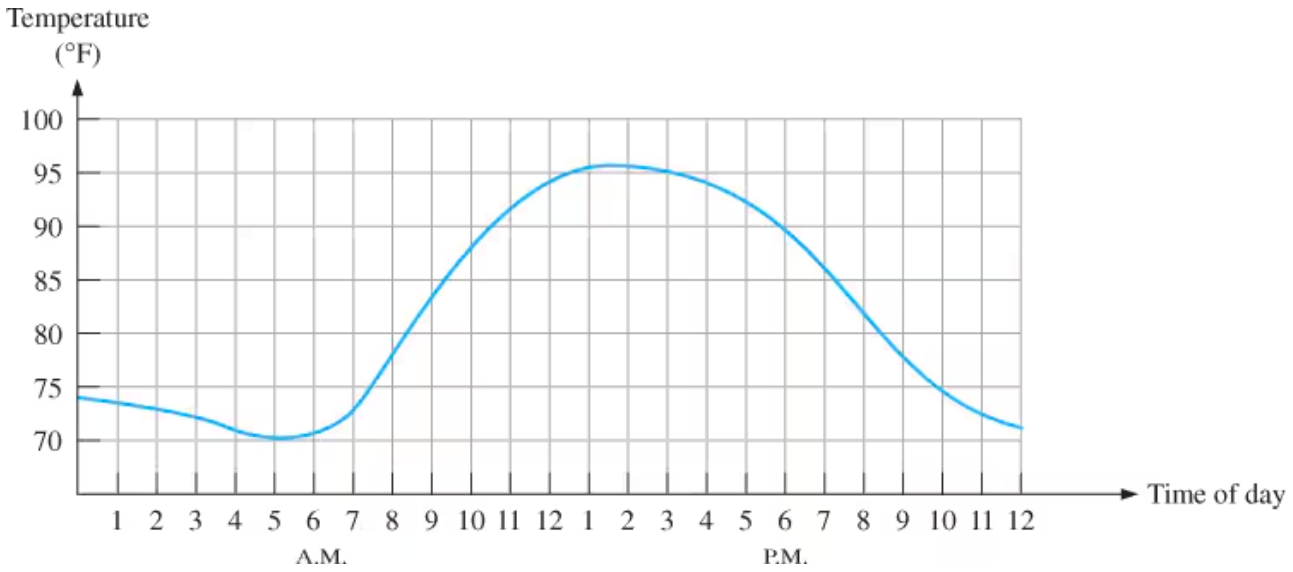
Slide Courtesy:
Digital Fundamentals, Eleventh Version, Thomas Floyd

Outline

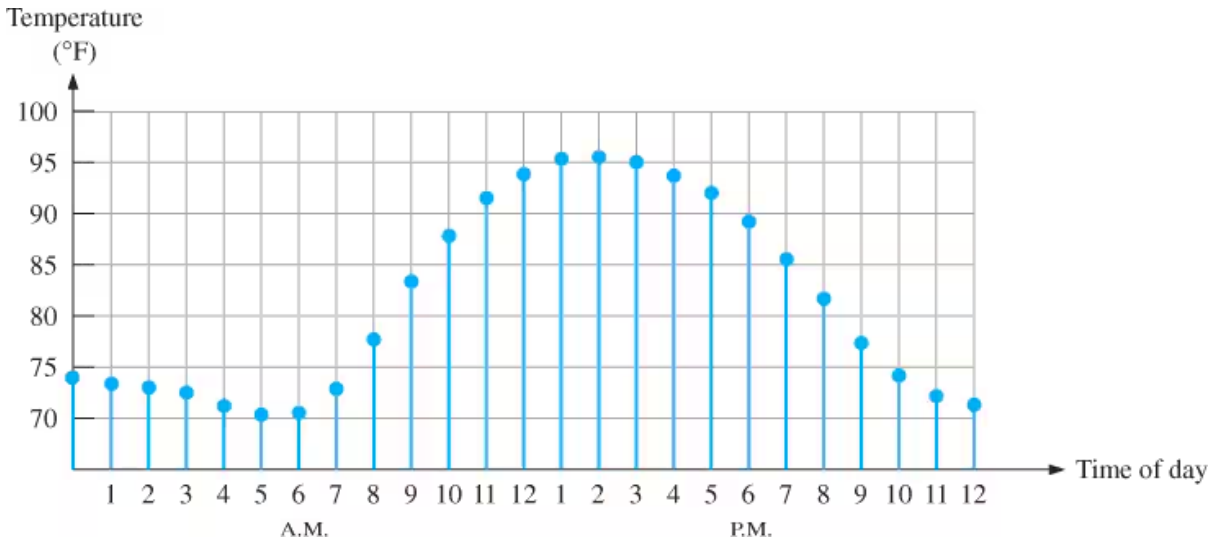
- Digital Vs Analog
- Binary Digits
- Logic Levels
- Basic Logic Functions and Logic Gates
- Implementation examples

Analog Vs Digital Systems

- **Analog:** Continuous signals (e.g., sound waves, voltage, temperature)
- **Digital:** Discrete signals (e.g., 0s and 1s)
- Advantages of Digital: Noise immunity, easier storage, processing



Graph of an analog quantity (temperature versus time).



Sampled-value representation (quantization) of the analog quantity

Binary Digits

Binary Digits (Bits):

- Each of the two digits in the binary system, 1 and 0, is called a bit.
- Each binary digit (bit) represents an **electrical state**:
  - **0** = Low voltage
  - **1** = High voltage
- Bits: Smallest unit of digital information

Binary Numbers System

- Binary: Base-2 system (0 and 1)
- The two binary digits (bits) are 1 and 0.
- The position of a 1 or 0 in a binary number indicates its weight, or value within the number, just as the position of a decimal digit determines the value of that digit.

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

In general, with n bits you can count up to a number equal to $\mathbf{2^n-1}$.

Conversions

Binary to Decimal Conversion:

- The decimal value of any binary number can be found by adding the weights of all bits that are 1 and discarding the weights of all bits that are 0.

Q. Convert the binary whole number 1101101 to decimal.

Weight:	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Binary number:	1	1	0	1	1	0	1

$$1101101 = 2^6 + 2^5 + 2^3 + 2^2 + 2^0$$
$$= 64 + 32 + 8 + 4 + 1 = \mathbf{109}$$

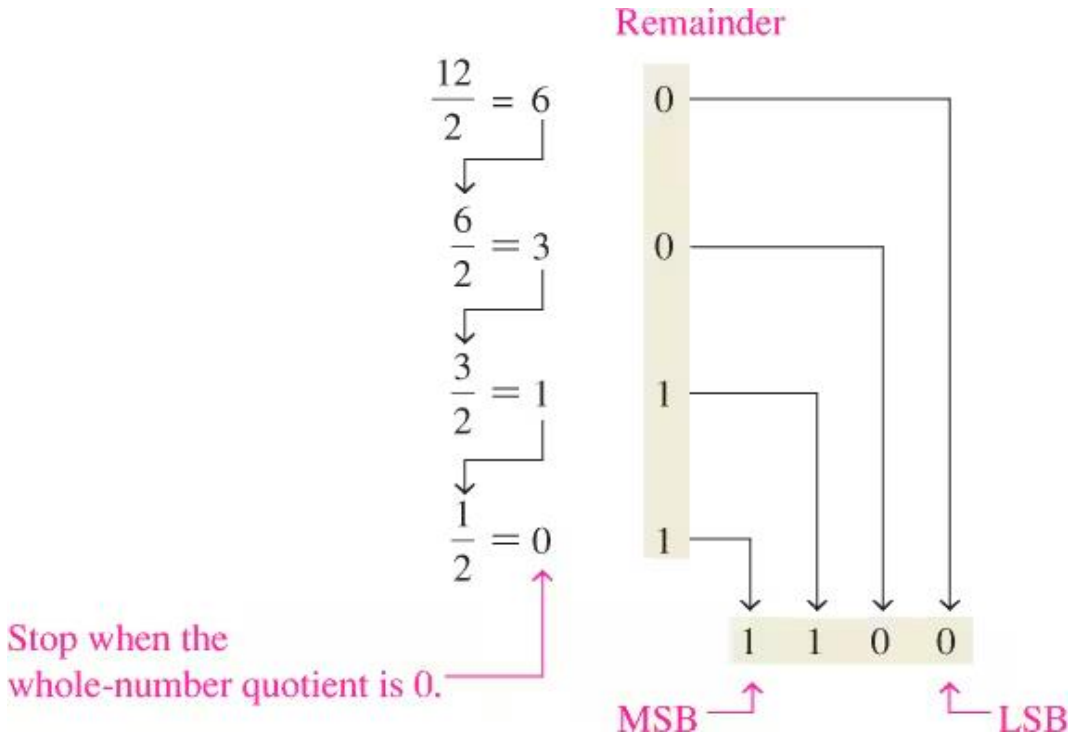
Q. Convert the fractional binary number 0.1011 to decimal.

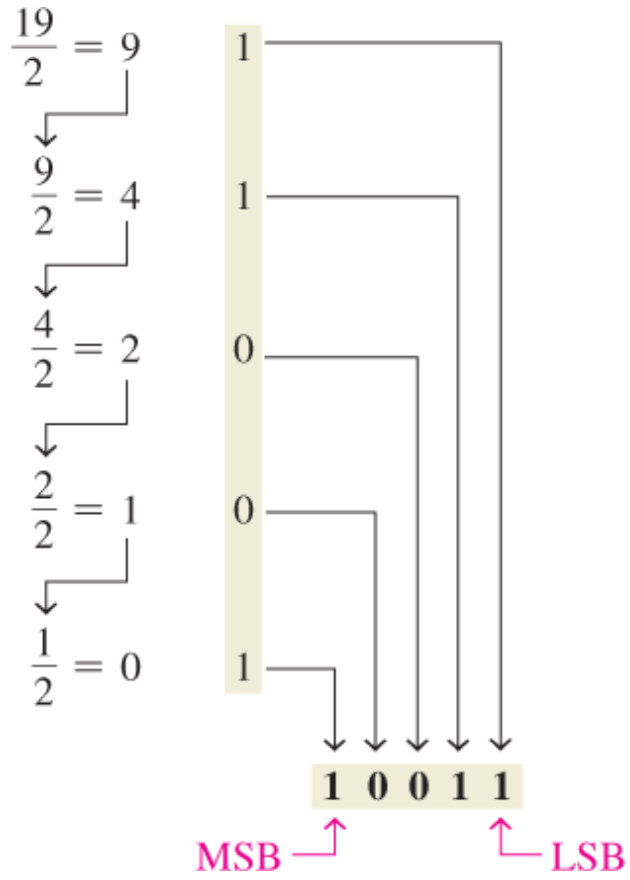
Weight:		2^{-1}	2^{-2}	2^{-3}	2^{-4}
Binary number:	0.	1	0	1	1

$$0.1011 = 2^{-1} + 2^{-3} + 2^{-4}$$
$$= 0.5 + 0.125 + 0.0625 = \mathbf{0.6875}$$

Conversions: Decimal to Binary Conversion

- A systematic method of converting whole numbers from decimal to binary is the **repeated division-by-2 process**.



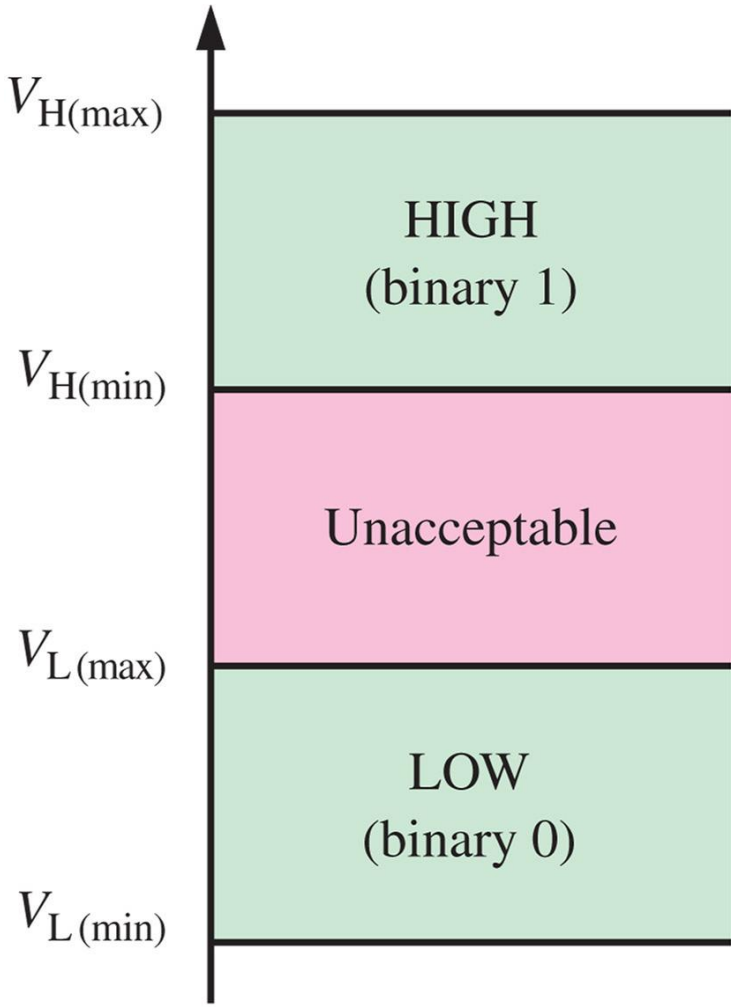


Logic Levels

Logic 0: Low voltage (e.g., 0V)
Logic 1: High voltage (e.g., 5V or 3.3V)

- The voltages used to represent a 1 and a 0 are called logic levels.
- In a practical digital circuit, a HIGH/LOW can be any voltage between a specified minimum value and a specified maximum value.

- Example:
  - 2V to 3.3 V → HIGH (binary 1);
  - 0 to 0.8 V → Low (binary 0) ;
  - Voltages between 0.8 V and 2V are unacceptable.



Logic level ranges of voltage for a digital circuit

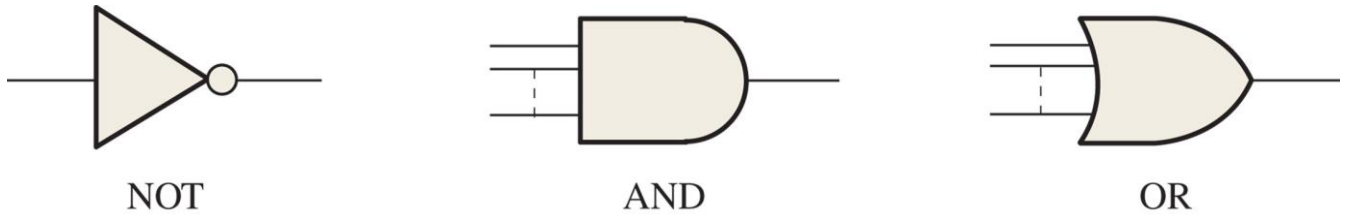
Basic Logic Functions

- AND: True only if all inputs are true
- OR: True if at least one input is true
- NOT: Inverts the input (0 becomes 1, 1 becomes 0)

Logic Gates

- AND
- OR
- NOT
- NAND
- NOR
- X-OR
- X-NOR

- Logic gates are the physical building blocks of digital circuits and are found in most electronic devices.
- A circuit that performs a specified logic function is called a logic gate.
- A logic gate performs a logical operation on binary inputs to produce a binary output either true (1) or false (0).



The basic logic gates symbols.

AND Gate

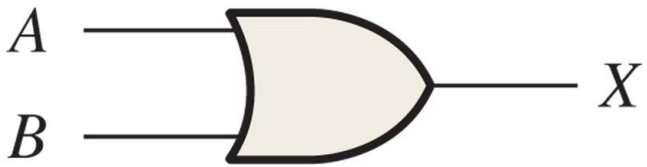


Truth table for a 2-input AND gate.

Inputs		Output
A	**B**	**X**
0	0	0
0	1	0
1	0	0
1	1	1

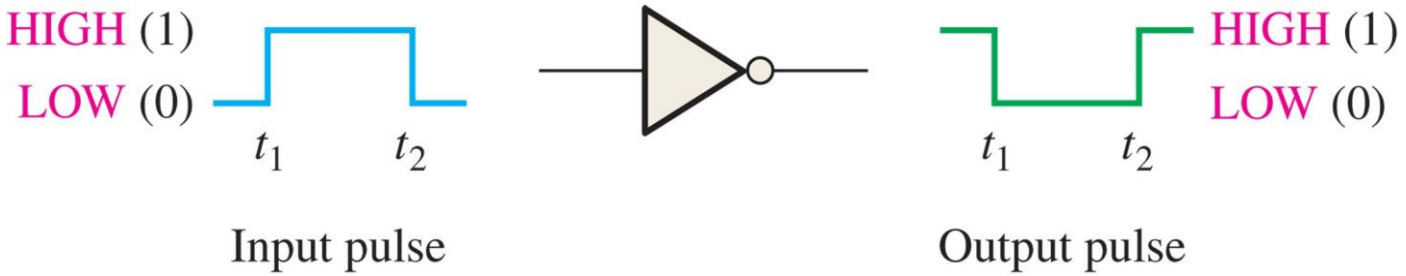
1 = HIGH, 0 = LOW

OR Gate



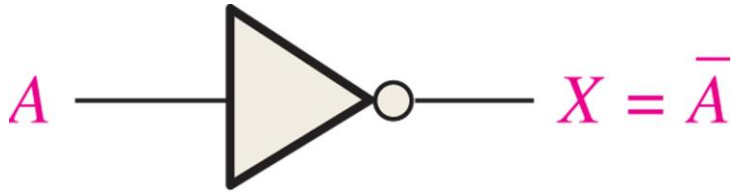
A	B	$A + B = X$
0	0	$0 + 0 = 0$
0	1	$0 + 1 = 1$
1	0	$1 + 0 = 1$
1	1	$1 + 1 = 1$

Inverter (NOT Gate)

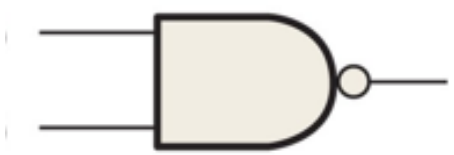


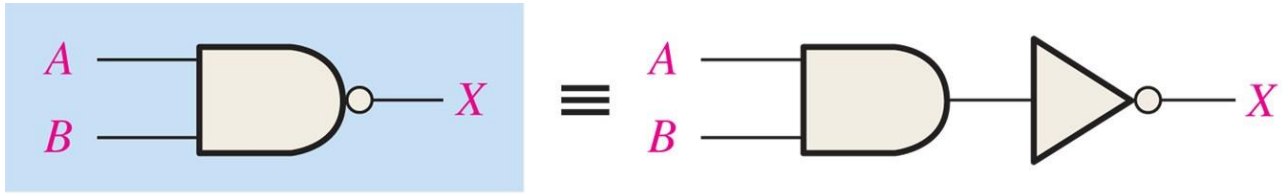
Inverter truth table.

Input	Output
LOW (0)	HIGH (1)
HIGH (1)	LOW (0)



NAND Gate





(a) Distinctive shape, 2-input NAND gate and its NOT/AND equivalent

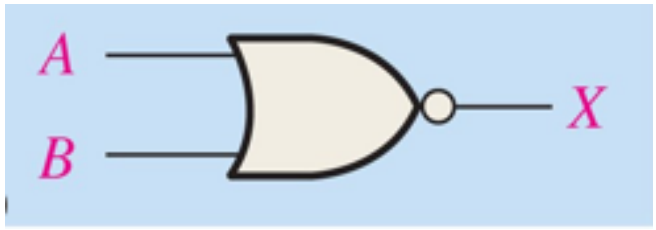
Truth table for a 2-input NAND gate.

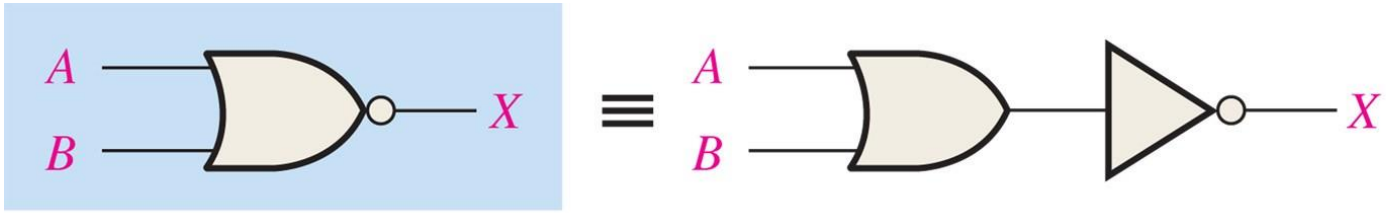
Inputs		Output
A	*B*	*X*
0	0	1
0	1	1
1	0	1
1	1	0

1 = HIGH, 0 = LOW.

Opposite of AND

NOR gate





(a) Distinctive shape, 2-input NOR gate and its NOT/OR equivalent

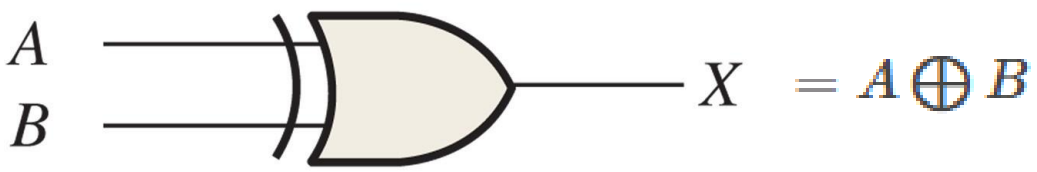
Truth table for a 2-input NOR gate.

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	0

1 = HIGH, 0 = LOW.

Opposite of OR

Exclusive- OR (XOR) gate



⊕: Denotes the XOR operation

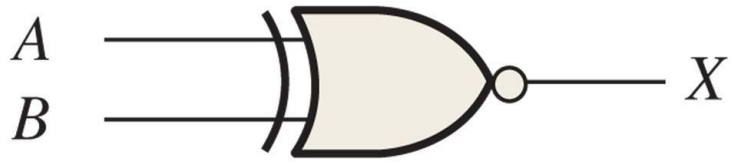
$$A \oplus B = A\bar{B} + \bar{A}B$$

Truth table for an exclusive-OR gate.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	0

Output is 1 when inputs differ

Exclusive- NOR (XNOR) gate



$$= (\overline{A \oplus B}) = (A.B + \overline{A}.\overline{B})$$

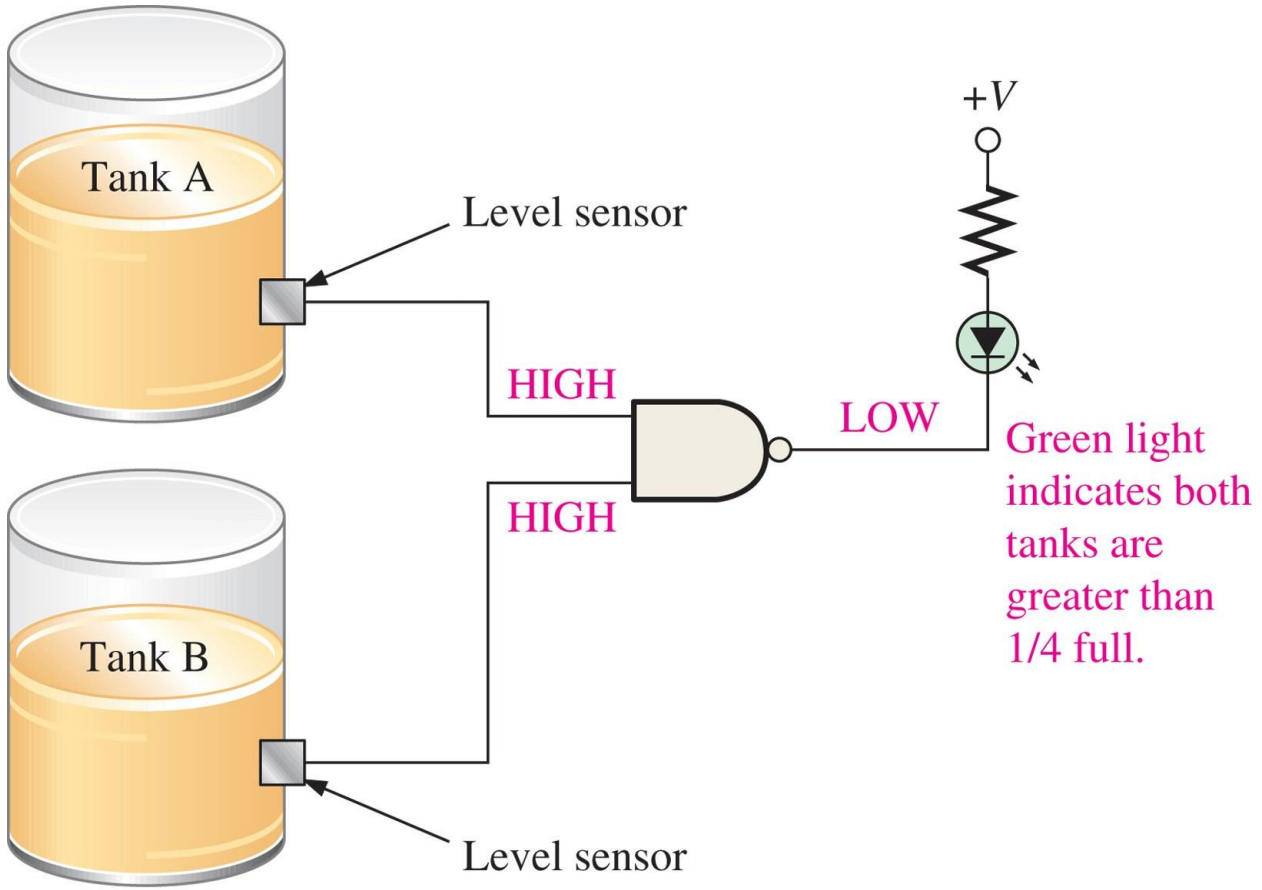
Truth table for an exclusive-NOR gate.

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	1

- Output is 1 when inputs match

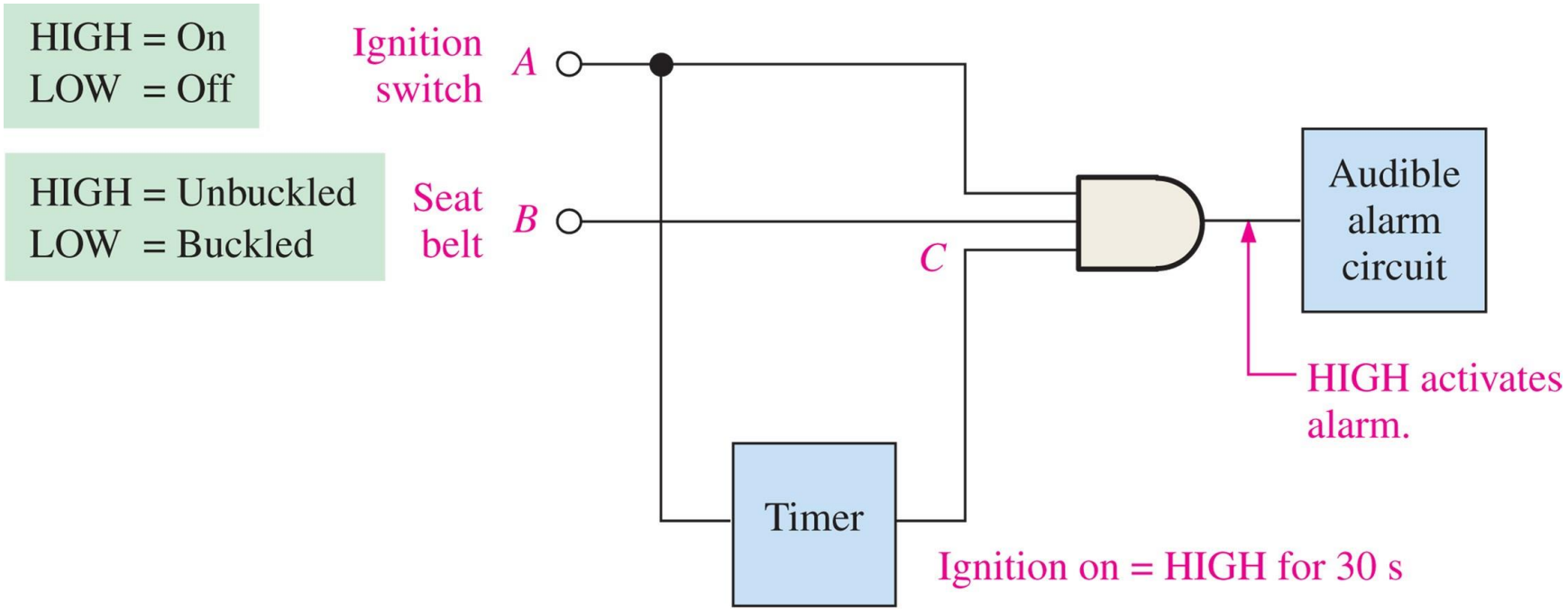
Tank Level Indicator using NAND gate

- If tank A and tank B are above one-quarter full, the LED is on.

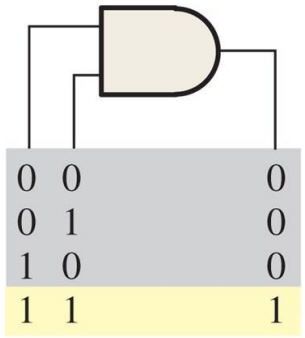


Example implementations

A simple seat belt alarm circuit using an AND gate.

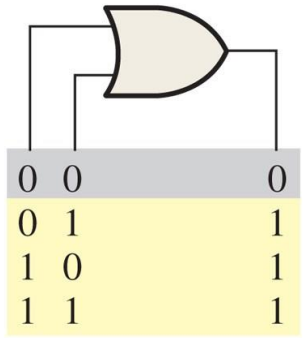


Logic Gates Summary



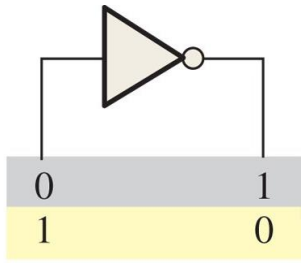
0 0	0
0 1	0
1 0	0
1 1	1

AND



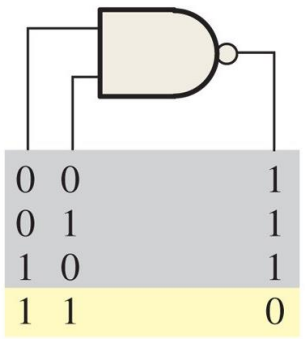
0 0	0
0 1	1
1 0	1
1 1	1

OR



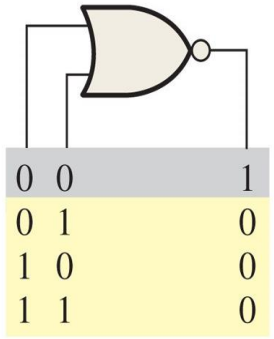
0	1
1	0

Inverter



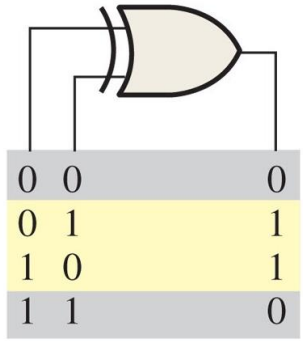
0 0	1
0 1	1
1 0	1
1 1	0

NAND



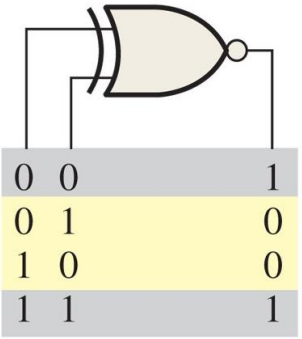
0 0	1
0 1	0
1 0	0
1 1	0

NOR



0 0	0
0 1	1
1 0	1
1 1	0

Exclusive-OR



0 0	1
0 1	0
1 0	0
1 1	1

Exclusive-NOR

Note: Active states are in yellow color

Summary

- **Analog signals** are continuous and need to be converted to digital signals (1's and 0's) for digital processing.
- **Binary numbers** are the language of digital electronics, forming the basis of data representation.
- **Logic gates** process binary data through simple operations to perform complex functions in digital circuits.